

Faithfulness of the target-state GNS representation

MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective

theorem

The ideal dichotomy of the same target forces the constructed nonzero representation to be injective.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. Choose the state extension $\phi : T \rightarrow \mathbb{C}$ of τ supplied by state extension along ι . Its GNS representation is $(H_\phi, \rho_\phi, \xi_\phi)$, with

$$\|\xi_\phi\| = 1, \quad \phi(t) = \langle \xi_\phi, \rho_\phi(t)\xi_\phi \rangle.$$

The inner product is linear in its second argument. The T -orbit of ξ_ϕ is dense by the GNS construction. The representation $\rho_\phi : T \rightarrow B(H_\phi)$ is injective.

Assumptions

The representative shell family is fixed. The established closed two-sided ideal dichotomy of this same T is available as a theorem. The GNS space is nonzero because its canonical vector has norm 1.

Conclusion

If $\rho_\phi(t) = \rho_\phi(s)$, then $t = s$. Thus the GNS representation of the chosen state on T is faithful as an algebra representation.

This does not infer faithfulness of an arbitrary vector state from faithfulness of its representation. State faithfulness requires its own argument. No irreducibility of ρ_ϕ , purity of ϕ , or separability of T is assumed.

Proof route

Apply the closed-ideal dichotomy to the kernel and exclude the whole-algebra case using the unit on the nonzero GNS space.

▼ Proof steps

1. The kernel $I = \ker \rho_\phi$ is a norm-closed two-sided ideal. The previously established result `shellFamilyTarget_closedIdeal_dichotomy` gives $I = \{0\}$ or $I = T$.

2. If $I = T$, then $\rho_\phi(1) = 0$. But ρ_ϕ is unital, so

$$\xi_\phi = I_{H_\phi} \xi_\phi = \rho_\phi(1) \xi_\phi = 0,$$

contradicting $\|\xi_\phi\| = 1$. Hence $I = \{0\}$, which is precisely injectivity. The source applies `Representation.injective_of_closed_ideal_dichotomy` after installing the separately proved nontriviality of H_ϕ .

Main citations

- The stated existence or structural result — `MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective`
- The fixed concrete target and source map — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- The target-state GNS definition — `MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation`
- Nontriviality witnessed by the unit vector — `MathlibAnnex.CStarAlgebra.CAR.nontrivial_tracialHilbertSpace`
- Closed-ideal dichotomy of the fixed target — `MathlibAnnex.CStarAlgebra.CAR.shellFamilyTarget_closedIdeal_dichotomy`
- The kernel argument for a nonzero unital representation — `MathlibAnnex.Analysis.CStarAlgebra.Representation.injective_of_closed_ideal_dichotomy`

Supporting route explanation

Lean source signature (exact)

```
theorem tracialRepresentation_injective (family : RepresentativeShellFamily) :
  Function.Injective (tracialRepresentation family)
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily</code>	The fixed shell family and resulting target T , source map ι and chosen trace extension ϕ .
<code>tracialRepresentation family</code>	The unital GNS action $\rho_\phi : T \rightarrow B(H_\phi)$ of ϕ on its complete GNS space, with canonical vector ξ_ϕ of norm one.
<code>Function.Injective (tracialRepresentation family)</code>	If $\rho_\phi(t) = \rho_\phi(s)$ as bounded operators on H_ϕ , then $t = s$ in T . This is faithfulness of the representation, not irreducibility or a direct assertion of state faithfulness.

Further source notes: In the linked proof, `nontrivial_tracialHilbertSpace` supplies a nonzero codomain, and `shellFamilyTarget_closedIdeal_dichotomy` is passed to the generic faithful-representation lemma. Neither is a new assumption added to the displayed signature.

▼ **Exact content identity**

Declaration: MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective

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